No False Green: A Benchmark for Evidence-Aware Compliance Verdicts over Structured Employer Records

Status of this draft: complete, for review. All sections are written. Every table is generated from retained artifacts by the repository's scripts, and every quantitative claim in the text is recomputed from those artifacts by `scripts/verify_paper.py`, which currently confirms 149 of 149. Each methodological rule that could have been chosen after seeing results (label views, model substitutions, evidence classes, excerpt selection, label audit, cut-offs) was committed to the repository before the results it governs, and the paper cites the record.

Abstract

Automated systems increasingly issue machine-generated compliance verdicts in regulated domains, presenting an organisation's obligations as a dashboard of satisfied and violated states. The evaluation of such systems measures accuracy on isolated determinations, which leaves the deployment-critical error unmeasured: reporting an obligation as *satisfied* when the evidence required to assess it was never present. We call this a **false green**, and we argue it is categorically worse than a false alarm, because a false alarm is corrected by review whereas a false green suppresses the review that would correct it.

We introduce **No False Green**, a benchmark for evidence-aware compliance verdicts over structured employer records in United States employment-based immigration. An instance is a synthetic employer: a bundle of source exports paired with a rulebook of 194 citable obligations across ten regulatory regimes. A system must return one of four verdicts for each (obligation, entity) pair: `pass`, `fail`, `needs_evidence`, or `unmonitored`. Ground truth is fixed by construction. A generator selects each pair's disposition first, then renders source records consistent with that choice, so labels are not annotations of data but the specification the data was rendered from.

The benchmark inverts the incentive structure of standard evaluation in one specific way: correct abstention is the **majority class**, at 201 of 221 pairs in the hand-authored anchor firm and 45.3% of 58,961 pairs across the released set. We report the false-green rate alongside violation recall, false-alarm rate on planted controls, abstention precision and recall, and a consistency check between a system's headline counts and its own per-pair verdicts. Trivial baselines establish that these metrics are not individually sufficient: a system answering `pass` everywhere attains a 0.000 false-alarm rate and is indistinguishable from a safe system under accuracy-style scoring, while scoring 1.000 on the false-green rate; conversely a system abstaining everywhere attains 0.632 abstention F1 while detecting no violations. An evidence-aware oracle attains 1.000 on every axis, establishing that the task is solvable and that low scores reflect the system under test.

We evaluate six free API models under two output schemas on a 140-pair declared subset. Offered an explicit abstention, they report 0.042 to 0.142 of unassessable obligations as satisfied, but detect only 0.050 to 0.450 of violations. With the abstention removed from the schema, false greens rise three- to fivefold, to 0.175 to 0.525, though prompt, data and models are unchanged. False greens concentrate where an export carries a populated column adjacent to the missing evidence, such as a signed date standing in for a signature. Where nothing adjacent is exported, models raise no false green at all. Because every instance is synthetic, the benchmark is released without the confidentiality constraints that keep comparable compliance data unpublishable.

---

1. Introduction

A compliance dashboard makes a claim about the world with every cell it renders. When a cell is green, it asserts that a specific legal obligation, attached to a specific worker or filing, has been assessed and found satisfied. The assertion has a precondition that the rendering does not express: that the evidence needed to perform the assessment was present.

When the precondition fails, a system has three options. It can say so. It can decline to render the cell. Or it can render the cell green. The third option is the one this paper is about, and it is attractive to build for reasons that have nothing to do with correctness: a dashboard of mostly-green cells looks like a working product, a mostly-grey one looks broken, and the customer cannot distinguish "assessed and fine" from "unassessed" without reading the underlying records themselves, which is the labour the product exists to remove.

We call the third option a **false green** and make three claims about it.

It is not symmetric with a false alarm. A false alarm consumes reviewer time and is extinguished by the review it triggers. A false green consumes nothing and triggers nothing. It is self-concealing: the only signal that would surface it is the review that it suppresses. In immigration compliance specifically, the obligations at issue carry per-worker civil penalties and the enforcement mechanism is an audit that arrives without warning, so a false green converts a curable gap into an uncurable finding.

It is invisible to accuracy. Violations are a minority class in real compliance data. A system that answers `pass` on every pair whose evidence is absent therefore scores well on any metric that averages over pairs, and scores *perfectly* on the compliant cases that dominate the distribution. Section 8 shows this concretely: our `always_pass` baseline attains a false-alarm rate of 0.000 across 31 firms.

It is not measured. Benchmarks in legal NLP evaluate reasoning and classification over regulatory text and case law. Benchmarks for abstention evaluate free-text question answering, where "insufficient evidence" is a property of a query. Neither addresses a system reading an organisation's structured records and deciding, per obligation, whether the records support a verdict at all.

We make three contributions.

1. **A task.** Verdicts over (obligation, entity) pairs with four states, where `needs_evidence` is a first-class output with its own correctness condition rather than a refusal to participate (Section 4).
2. **A benchmark.** A hand-authored anchor firm with 221 pinned expectations, a seeded generator producing unbounded additional firms with ground truth by construction, and a rulebook of 194 obligations each carrying its regulatory citation (Section 5).
3. **A metric suite** centred on the false-green rate, with baselines demonstrating that no single metric in the suite is sufficient alone (Sections 6 and 8).

The benchmark is fully synthetic, which is the reason it can exist. Compliance data describing real employers is subject to attorney-client privilege and employee privacy obligations that make publication impossible, which is why the gap this benchmark fills has remained open.

---

2. Related work

Four literatures touch this problem. None occupies the position this benchmark takes, and the reason in each case is a difference in what the system is given and what it must return.

Legal and regulatory NLP benchmarks. Work in this area evaluates reasoning and classification over *regulatory text*: statutes, case law, policy manuals. A system is given a legal question or a document and asked to answer, classify, or retrieve. Our input is an organisation's structured operational records, and our output is a verdict per obligation about that organisation. A model that scores well on legal reasoning has demonstrated nothing about whether it will report an unassessable duty as satisfied, because the task never asks it to decide whether its evidence base is sufficient.

Immigration-specific datasets. Recent work covers question answering over immigration policy sources, and reasoning over administrative appellate decisions including adjudicator error. Both are beneficiary-side and text-side:

the questions concern what the law says or how a decision was reasoned. Employer-side obligations, which is where the per-worker civil penalties and the audit exposure actually sit, are absent. We are not aware of a public dataset of employer immigration-compliance records with labelled obligation outcomes, and the reason is straightforward: real ones cannot be published.

Compliance benchmarks for AI systems. A growing body of work assesses whether an AI system complies with rules governing AI, including regulation-conformance assessment and runtime governance-evidence sufficiency for agent audit trails. The direction of assessment is inverted relative to ours. There, the system under test is the regulated subject. Here, the system under test is the *assessor*, and the regulated subject is a third party whose records it reads. The nearest neighbour in spirit is work asking whether an evidence bundle suffices to reconstruct a decision; the difference is that we ask whether a system correctly declines to issue a regulatory verdict about someone else.

Abstention and selective prediction. This is the closest literature and deserves the most precision, because a reviewer will reasonably ask what is new.

Selective prediction gives a model the option to abstain and evaluates it on a risk-coverage curve: coverage is a good to be maximised subject to a risk budget, and abstention is a *cost* incurred to avoid error. Recent abstention benchmarks evaluate whether models recognise unanswerable or underspecified questions, typically in free text, where insufficiency is a property of the query and ground truth is established by annotation.

Three things differ here.

First, abstention is not a cost. For 45.3% of pairs in our released set it is the *correct output*, and it is actionable: `needs_evidence` names a document the firm must produce. A system that trades abstention for coverage is not moving along a tradeoff curve, it is producing wrong answers that happen to look like answers.

Second, insufficiency is a property of the evidence base rather than the query, so it is determinable **by construction**. The generator withholds a required field and the label follows mechanically. Free-text abstention benchmarks must annotate insufficiency, and annotation of legal sufficiency is exactly the judgement call that makes such datasets small and contestable.

Third, the error is directional in a way the risk-coverage framing does not capture. Answering `fail` on an undecidable pair is wrong but triggers review. Answering `pass` is wrong and *suppresses* review. Aggregating both into a single risk term hides the only distinction that matters operationally, which is why we report the false-green rate as a separate quantity rather than as one minus abstention recall.

Aggregate consistency. We are not aware of prior work measuring whether a system's reported headline counts agree with its own per-item outputs. We include it as a metric because we have observed the failure in deployed compliance dashboards, where a card and its drilldown disagree; we flag it as the least-established axis in the suite.

Positioning, stated plainly. The contribution is not a new abstention algorithm and not a theoretical result. It is a task formulation in which correct abstention is the majority class and is fixed by construction, an instrument for measuring the directional error that matters in deployment, and a releasable artifact in a domain where releasability is the binding constraint. Section 10 states what this does not establish.

---

3. Background: the four states

The four-state vocabulary is not a design choice we made for the benchmark. It is forced by the professional-responsibility constraints on any system that reports on legal obligations without being counsel, and it was arrived at in a deployed product before it was formalised here.

`pass`**.** A rule was evaluated against present evidence and the evidence satisfies it. The claim must be time-bound, because a pass is only as current as the last data import.

`fail`**.** A rule was evaluated against present evidence and the evidence violates it. The claim must be framed as a potential issue requiring review rather than as a legal conclusion, because characterising conduct as unlawful is an act of legal judgement.

`needs_evidence`**.** The rule could not be evaluated because a required input is absent. This is the state that most often collapses in practice, and the collapse has a direction: a file that was never provided is reported as a file that is in order. The two are opposite. A not-provided document is an **open gap** that counts against audit readiness.

`unmonitored`**.** The obligation is not assessable from this class of source data at all. An employer export cannot evidence whether an adverse action was taken against a worker while a contested tentative nonconfirmation was pending, because that fact lives in conduct and not in a record the employer exports. Such an obligation must remain visible as unmonitored. Silent omission is itself a false green.

The distinction between `needs_evidence` and `unmonitored` matters operationally: the first is actionable by the firm (provide the document), the second is not (no document exists to provide), and conflating them either sends firms on impossible errands or hides real gaps.

---

4. The task

An instance is a firm $F = (S, E, R, a)$ where S is a set of source exports, E a set of entities, R a rulebook, and a a pinned evaluation date.

An **entity** $e in E$ has a type: `worker`, `lca`, `worksite`, `wage_cohort`, or `employer`. Entity type matters because obligations attach at different granularities: an I-9 timeliness duty attaches to a worker, a wage-floor duty to a filing, a prevailing-wage comparison to a cohort defined by occupation, level, and location, and a site-visit readiness duty to the employer as a whole.

An **obligation** $r in R$ carries a regulatory citation, the entity type it attaches to, and a set $mathrm{req}(r)$ of source fields that must be present for the obligation to be decidable. This last component is what makes `needs_evidence` computable rather than a judgement call: if a field in $mathrm{req}(r)$ is absent from S for entity e, the pair is not decidable, and any verdict other than `needs_evidence` is wrong regardless of whether the underlying situation happens to be compliant.

A **pair** is (r, e) where e's type matches r's attachment. The prediction target is a verdict $v in {texttt{pass}, texttt{fail}, texttt{needs_evidence}, texttt{unmonitored}}$, optionally with evidence pointers and a citation.

All date-relative obligations evaluate against a rather than against the current date, so planted expiry windows remain valid indefinitely. The anchor firm pins a = 2026-06-29.

Scoring rule. A pair with no prediction is counted as an error rather than skipped. A system cannot raise its score by declining to answer the pairs it finds hard. This is enforced in the harness and tested.

---

5. Benchmark construction

5.1 The rulebook

The rulebook contains 194 obligations across ten regimes: I-9 and E-Verify (28), Labor Condition Application (23), Public Access File (15), wage and prevailing wage (21), H-1B status (16), I-129 supplement (13), L-1 (21), F-1 OPT and STEM OPT (15), site-visit readiness (14), and cross-cutting deadlines and penalties (28). The source manifest lists 196 entries; two rules appear deliberately in two regimes, once as a field check and once as a deadline clock, and are counted once here under I-9.

Each obligation carries the check it performs, its threshold, its regulatory citation, and a description of the information needed to decide it. The citations had previously been through a verification pass in which incorrect citations and thresholds were corrected inline against primary sources. The rulebook is built programmatically from the source engine's machine-readable registry of that manifest rather than retyped, so every citation is traceable to a source document rather than to our recollection.

The anchor firm's expectation set also refers to 17 rules by the engine's internal identifiers rather than manifest identifiers. Fourteen map to the manifest rule stating the same duty through the engine's own published alias table. Three are engine roll-ups whose definitions are assembled from the manifest rules they aggregate and carry those rules' citations. No obligation text in the benchmark is written from scratch.

5.2 The anchor firm

The anchor firm is a synthetic employer of twelve workers, nine on H-1B status and three United States citizens, operating from two worksites, with eleven source exports totalling 93 data rows: HR roster, case-management export, certified LCAs, I-9 records, E-Verify cases, payroll register, RFE and NOID cases, cap cases, green-card cases, PERM recruitment tracker, and benefits census.

Each worker exercises one planted regulatory mechanism. The design is one clean plant per persona so that a missed finding localises to a mechanism rather than to an interaction. Representative plants:

Mechanism	Planted signal
I-9 Section 2 timeliness	Section 2 completed 8 business days after start, against a 3-day limit
Reverification deadline	List A document expiring 14 days after the pinned date
E-Verify TNC window	Notice delivered, no employee decision recorded, well past the response window
LCA expiration	Certified period ending 4 days after the pinned date
Required wage, greater-of	Salary equals the certified rate but falls below the prevailing wage
Material change, occupation	Filing carries an individual-contributor occupation code against a management title
Material change, worksite	Roster location differs from the filing's stated area of intended employment
Petition expiry	Validity ending 21 days after the pinned date

The firm also carries **clean controls**: one H-1B worker whose records conform on every dimension, and three United States citizens outside H-1B scope. A finding on a control is a false alarm the harness scores.

Its expectation set contains 221 pairs: 20 planted violations and 201 evidence requests. The 201 arise because most obligations in this domain require evidence an employer export does not contain. A certified LCA does not evidence that the required notice was posted; a roster does not evidence which document a worker presented. The honest output for such a pair is an evidence request, and the fact that these outnumber confirmed violations tenfold is the empirical claim underlying the benchmark's design.

5.3 The generator

The anchor firm is too small to support a confidence interval on any system's error rate. The generator addresses this.

Its central property is that **labels precede data**. For each applicable (obligation, entity) pair the generator draws a disposition:

- `planted_violation`, and a specific regulatory breach is rendered into the records;
- `planted_control`, and a conforming case is rendered, with a configurable share placed

exactly on the legal boundary (I-9 Section 2 on business day three of three, an LCA expiring on day eight of a seven-day window, a salary equal to the floor) to test whether a system flags on proximity rather than on the rule;

- `withheld_evidence`, and a required input is omitted;
- `not_applicable`, for obligations with no expression in export data.

Only after the full disposition map is drawn are the CSVs written. There is consequently no annotation step and no annotator disagreement to report, and each label is auditable back to the decision that produced it. The cost of this design is that the generator can only express mechanisms we have implemented renderers for; unrendered obligations reach the label set as withheld-evidence or not-applicable, which is honest but limits the diversity of planted violations relative to the hand-authored firm.

Generation is deterministic in the seed. Two runs of the same seed produce byte-identical label sets and firm metadata, which the test suite asserts.

5.4 Released scale

	Value
Firms	31 (1 anchor + 30 generated at seeds 1-30)
Pairs	58,961
Withheld-evidence pairs	26,682 (45.3%)
Planted violations	7,044
Obligations	194

Additional firms are available at any seed.

---

6. Metrics

Let T be ground truth and P predictions over pairs.

Violation recall. Of pairs labelled `fail`, the share predicted `fail`. A missed violation is the failure the domain treats as unforgivable, so this is reported first.

Violation precision and **false-alarm rate.** Precision over `fail` predictions, and separately the share of `planted_control` pairs on which a system raised any alarm. The second is reported separately because controls are the pairs where a false alarm is unambiguously an error.

False-green rate. The headline metric:

mathrm{FGR} = frac{|{p : T(p) = texttt{needs_evidence} wedge P(p) = texttt{pass}}|}{|{p : T(p) = texttt{needs_evidence}}|}

The share of undecidable pairs a system reports as satisfied. Note what it is not: it is not one minus abstention recall. A system that answers `fail` on an undecidable pair is wrong, but it is wrong in a direction that triggers review. Only `pass` suppresses review, so only `pass` counts here.

Abstention precision and recall, treating `needs_evidence` as a class. Precision matters because a system can drive the false-green rate to zero by abstaining everywhere, which the baselines demonstrate.

Evidence-pointer accuracy. Among pairs a system got right and for which ground truth records the relevant fields, the share where the system's cited evidence intersects them. A correct verdict reached by citing the wrong record is not a reliable verdict.

Consistency drift. Mean absolute relative disagreement between the counts a system displays for a surface and the per-pair verdicts the same system emitted. A card reading twelve above a drilldown returning three is internally inconsistent regardless of which number is correct. This axis exists because we have observed the failure in deployed dashboards and are not aware of it being measured elsewhere.

Reporting requirement. The false-green rate must be reported alongside violation recall. Neither is meaningful alone, and Section 8 shows why.

Two label views, declared before any main-run result was read. The anchor firm's labels were written as the acceptance gate for a compliance engine, where `fail` means "requires action". This benchmark's `fail` means "the records show a violation". Ten of the 20 planted violations sit between those meanings, so we score under two views, fixed in `nofalsegreen/labelviews.py` on 2026-09-24 with each affected pair named by identifier.

- **Engine view, the headline.** The labels as authored. It is the headline because it

existed before this benchmark did, so nothing decided here can have tuned it.

- **Strict view.** Five "actual-wage memo missing" violations become `needs_evidence`: no

export records the memo as absent, the evidence is simply not there, which is this benchmark's own definition of an evidence request. Five deadlines that had not yet passed on the evaluation date, an LCA and a petition near expiry, an open extension window, and an RFE and a NOID not yet due, are removed from scoring and reported separately as deadline alerts, listing the verdict each model gave. The remaining ten violations are breaches that had already occurred.

The strict view's violation recall has a denominator of ten, so a single pair moves it by 0.1. It is always reported beside the engine view, never alone. A model that answers `needs_evidence` on the unevidenced memos loses recall under the engine view and none under the strict view; the test suite asserts exactly that, so the difference between the two views on those pairs is a direct measure of how often a model declined to call a missing document a violation.

A third view, from a blind audit of the withheld labels. While reading early results we found a withheld label the exports contradict. `i9.section2.employer_signature` is labelled `needs_evidence` because the source engine never maps the column, yet `section2_signed_date` and `section2_examiner_name` are populated for every employee, and the prompt tells models to match columns by meaning. Because this was noticed in model output, the correction is constrained so that it cannot be steered by that output. The protocol (`docs/LABEL_AUDIT_PROTOCOL.md`) was committed before the audit ran. It covers all 120 withheld pairs, not the ones models got wrong. The auditor is a separate agent given only each pair's obligation text and the exports its batch was shown, with no model reply, score, or engine note. A label is overturned only if every required item of "information needed" is present for that entity; ambiguity keeps the label.

The audit overturned 2 of 120 labels, the Section 1 and Section 2 I-9 signatures, both on signed-date columns populated for all twelve employees. The auditor flagged the Section 1 call as the weaker: there is no separate signature flag, and a stricter reader could hold that a date is not a signature. In the **audited view** those pairs are removed from scoring, not relabelled, since an auditor's verdict is not a label fixed by construction. Removing pairs can only remove false greens, so every table reporting this view states how many pairs it scores, and the engine view remains the headline. Every audit decision is published with its reasoning (`docs/label_audit.json`).

7. Baselines

Trivial baselines (run). `always_pass` answers `pass` everywhere and is maximally unsafe. `always_abstain` answers `needs_evidence` everywhere: safe but useless. `always_fail` flags everything. `oracle` reads the label basis and decides exactly when evidence is present; it is not a competitive system but an upper bound establishing that the task is solvable.

Language-model baselines. The harness was tested against mock models with known failure signatures before any real call was made. The protocol is frozen in code (`baselines/prompts.py`, `PROMPT_VERSION` 1.1.0) so that the reported result cannot be a post-hoc selection of prompt:

- *Prompt.* Per firm, the source exports verbatim as CSV, the obligations needed by the

batch with their check, threshold, citation and the information needed to decide them, and the pairs to decide. Two reading rules apply identically in both conditions: export columns are matched to obligations by meaning, and reference knowledge such as federal holidays and regulatory thresholds is supplied by the model rather than required from the exports. Verdicts are returned one per line as `pair_id`, verdict, evidence field. The evaluation date is pinned in the prompt with an explicit instruction not to use the real current date, since every deadline rule depends on it.

- *Output schema ablation.* Each model is run twice. In `abstain_offered` the schema

includes `needs_evidence` and states the `pass`-versus-`needs_evidence` distinction explicitly. In `abstain_withheld` only `pass` and `fail` are permitted. The second condition is not a strawman: it is what a system built to fill a dashboard looks like, where every cell must resolve and there is no representation for "we could not tell". The difference isolates how much of the false-green rate is an interface artifact rather than a reasoning failure. We separately count **off-schema** replies, where a model abstains despite the schema forbidding it, which we read as a finding rather than an error.

- *Models.* The protocol originally called for eight to twelve current instruction-tuned

models across providers and sizes. With no compute budget, the study ran six free API models from four providers (Section 7.2); results speak to those models only.

- *Aggregation.* Scored per firm, then averaged across firms with standard deviation.

Pairs within a firm share source records and are not independent.

Parsing policy, which is a scoring decision. A reply line naming a pair but no permitted verdict yields **no prediction**, and the metric module counts the pair as an error. It is never coerced. Coercing to `needs_evidence` would convert a model's formatting failures into abstentions and flatter its false-green rate; coercing to `pass` would do the reverse. Either would make the headline number a function of our parser. We report the unparseable rate separately so a reader can see how much of a score is formatting.

Coverage, because an unanswered pair cannot be a false green. A pair with no prediction scores as a miss on violation recall, which is conservative. On the false-green rate it is the opposite: the pair stays in the denominator and can never be counted as `pass`, so a model that answers less looks safer, and one that answers nothing scores a perfect 0.000. The headline rate keeps the declared definition, but every results table also reports how many pairs each model answered and the false-green rate among withheld pairs it actually answered. A gap between the two columns is a coverage effect, not safety.

Delivery failures are not answers. Free endpoints sometimes return an empty reply, end a reply with an upstream error, or cut it off below our output cap. None of these is the model's answer, so the batch is re-requested, at most three attempts in all. Three such failures in a row, across batches, means the host is failing rather than the batch: the run stops, keeps its complete batches, and resumes after the next quota reset. A run is never recorded as finished while a batch is missing for this reason, because its missing pairs would score as model errors when the cause was delivery. Two outcomes are the model's own and are kept as given: a reply that ends normally but contains no usable verdict lines (counted as unanswered), and a reply that runs to our 32,000-token cap on all three attempts (the cap sits at the evaluated models' own ceiling, so the model is generating without end; the batch is recorded as unanswered). Every attempt, with its token count and the provider's stop reason, is in the sweep log.

Two departures from this rule occurred, and both are disclosed here. First, Lightning's second run never finished normally: its host repeatedly cut replies off at a ten-minute limit. Under a cut-off rule committed before the final quota window, once every other run had finished, the run was given one pass that tried each remaining batch up to three times

and was then recorded with what it had answered: 68 of 140 pairs. Second, during that pass one reply ended with an upstream error *after* delivering all twelve verdict lines. The harness accepted it because the reply was not empty, contrary to the rule above. We keep it, since its answer is complete, and the harness has since been corrected to re-request such replies. It is the only reply in the study that ended without a normal stop.

Measured cost. The full sweep is 1,446 API calls per model per variant, about 2.55M input and 1.73M output tokens over 31 firms at batch size 40. At list prices that is roughly $9 to $170 per model per variant depending on tier, and $180 to $3,400 for ten models across both variants. Prompt caching on the sources-and-rulebook prefix should reduce input cost materially, since every batch within a firm re-sends the same prefix. The anchor firm alone costs under $1.50 on the most expensive tier, which is the recommended first run.

7.1 A pilot run, and the protocol revision it forced

The protocol above is version 1.1. Version 1.0 was run first, on the same 140 pairs, and completed four of twelve model-condition runs before we stopped it. We report what it found because it changed the protocol, and a reader should be able to judge whether the change was principled.

The pilot showed violation recall of 0.000 for both models whenever the abstention option was offered. On inspection this was mostly the harness, for three reasons. Thirty of the 140 pairs, including 10 of the 20 planted violations, used rule identifiers absent from the version 1.0 rulebook and reached the prompt with no obligation text at all. The evidence list shown for every obligation had been damaged by our extraction into run-together tokens that matched no export column, and it named reference data no export contains, such as a federal-holiday calendar. And version 1.0 defined `needs_evidence` as a required field being absent, so a model that followed the prompt literally was obliged to abstain. In the clearest case, a Section 2 completed eight business days late, both dates were in the I-9 export, and the model still answered `needs_evidence`.

Version 1.1 fixes these defects and changes nothing else. Every pair's rule is defined. The evidence description is the manifest's own prose. `needs_evidence` is defined by the firm's records rather than by field names. The same two reading rules are added to both ablation conditions. The verdict menus are otherwise unchanged. The evaluated pairs are identical, fixed in a file before either version ran, so any difference between the pilot and the main run is attributable to the prompt. The pilot's raw replies and scores are retained in the repository.

The first attempt at the version 1.1 run exposed one further defect, in the harness rather than the prompt. We had capped each reply at 8,400 output tokens. Reasoning models spend most of a reply thinking before they answer, and 46 of the first 84 batch replies were cut off or came back empty before any verdict was written. The pairs in those batches would have scored as misses, depressing recall for reasons that had nothing to do with the models. We raised the cap to 32,000 tokens, which is below every evaluated model's own limit, and re-requested only the affected batches. The 38 replies that had finished within the lower cap were kept, because a cap that was never reached did not affect them. Truncated or empty replies are no longer cached, and the parser now counts a pair as unparseable only if it is never answered, not each time a model mentions it while reasoning. The affected attempt is archived intact beside the pilot.

One pilot observation did not depend on the defects. When forced to choose between `pass` and `fail`, both pilot models called most unassessable obligations violations rather than compliant, with false-green rates of 0.117 and 0.075. We report version 1.1 as the result and the pilot only as the account of how the protocol reached its final form.

7.2 Running on free API tiers

We report results obtained entirely on providers' free tiers, which constrains the experiment in two ways that must be stated because both affect the measurement.

Source scoping. Free tiers cap tokens per minute, and the cap is checked against the largest single call, not the average. Sending all eleven exports on every call puts the anchor firm's largest call at 12,185 tokens at batch size 20, and 16,832 at batch size 40, which exceeds several free per-minute quotas outright. We therefore send only the exports whose

regime an obligation can be evaluated against, with the HR roster always included because every other source joins to it. This reduces the largest call to 10,688 tokens at batch size 20. Note that shrinking the batch does not help: the obligation text is re-sent with every call, so a smaller batch raises the per-call floor rather than lowering it.

Scoping is a change to the prompt, so we treat it as part of the protocol rather than an implementation detail. Two safeguards keep it from manufacturing abstentions. First, the prompt names the exports that were withheld and instructs the model not to infer that an absent field exists in an export it cannot see; a model that abstained merely because it suspected hidden data would otherwise inflate the apparent abstention rate. Second, an obligation whose regime is unclassified falls back to all sources, so the mapping can only narrow prompts for regimes it knows about and can never hide data from an obligation it has not classified. The test suite asserts that a correct model scores identically with scoping on and off, and that the pass-guessing baseline is still caught at a false-green rate of 1.000 under scoping.

Model coverage, and the limitation it creates. Free tiers offer a narrower and generally smaller set of models than the specification in Section 7 assumes. Any result we obtain this way therefore speaks to the models we could reach, not to the frontier. We state the models and their published quotas alongside every number and do not generalise beyond them. This is a genuine weakness of the study rather than a presentational choice, and it is repeated in Section 10.

A mid-study substitution, declared before its results. The study was declared with six free models. After four of the twelve model-condition runs had completed, OpenRouter withdrew the free versions of both Nex models (N2.5 Pro and N2.5 Mini); the paid versions remain, and paying for them is outside this study's zero-cost constraint. Three batches of Nex N2.5 Pro replies had been collected; they cover too few pairs to score and are reported as an incomplete run, not as a result. Dropping both models would have left four, three of them from one provider family (NVIDIA Nemotron). We therefore replaced them with Qwen 3.8 27B and Gemma 4 31B, chosen for provider diversity among the models still free whose output limit exceeds the 32,000-token cap, with Inkling named in advance as the substitute should either fail the same two-pair screen the original models passed. Inkling then turned out to refuse API calls outright (it is served only inside agent applications), so before any replacement reply existed the fallback was changed to dots.3 Note Preview, the one remaining free model from a provider family not already in the set. A transient upstream rate limit is not treated as a screen failure. Both replacements then returned upstream rate limits on every screen attempt for over an hour, so a cut-off was added, again before any replacement had answered a benchmark batch: when the sweep reaches a replacement it screens it once more, and a model that has still never passed yields its slot to the fallback. The fallback fills at most one slot; a second unscreenable replacement is dropped and reported as such. Every screen attempt is logged (`docs/replacement_screen_2026-09-30.json`). The replacements and the fallback rule were committed on 30 September 2026, before any reply from either model existed; as with the label views, the commit history is the evidence of that ordering. The replacements run under the identical prompt, subset, batch size, and output cap.

The declared evaluation subset. Evaluating all 221 anchor pairs costs 28 calls per model per condition. At a free ceiling of 50 calls per day, one model consumes more than half a day and a six-model, two-condition study takes a week. We therefore evaluate a **declared stratified subset** of 140 pairs, fixed by a seed recorded in the code (`nofalsegreen/subset.py`) before any model was scored.

The subset is constructed to preserve the properties the benchmark's claims rest on, and the test suite asserts each one:

- **every planted violation is included**, so violation recall over the subset is recall

over the firm's complete violation set rather than a sample of it;

- **the class ratio is held at roughly 6:1** in favour of abstention. Spending the whole

budget on additional violations would have raised the subset to a 3.8:1 ratio and made the task easier than the firm it represents, which would flatter any false-green rate measured on it;

- **every regulatory regime present in the firm keeps at least one pair**, so the

per-regime table has no silently missing rows;

- **selection is deterministic**: the same seed yields the same 140 pairs permanently.

The cost is a smaller denominator, so the confidence interval on a subset false-green rate is wider than a full-firm rate would give. We report the subset size beside every subset result and never present a subset rate as a full-firm rate. The trivial baselines behave identically on the subset, with `always_pass` still scoring 1.000 on the false-green rate and the oracle still scoring 1.000 recall, so the subset does not make the task easier in the direction that matters.

Throughput. At a representative free quota of 10 requests per minute and 250 per day, the anchor firm across both ablation conditions is 24 calls, or about a tenth of one day. Adding five generated firms is roughly four days of quota. The runner paces itself under the per-minute limit, backs off on a transient rate-limit response, stops cleanly when the daily ceiling is reached, and resumes from a checkpoint, so a multi-day run is a matter of re-issuing the same command. Every model response is retained so that any reported number can be traced back to output.

A baseline we deliberately exclude. The anchor firm's expectation set originated as the acceptance gate for a commercial compliance engine. That engine would score near ceiling on this firm for a reason that has nothing to do with capability: the expectations and the engine were developed against each other. Reporting it as a baseline would be misleading, and we do not.

---

8. Results

8.1 Trivial baselines

Table 1 — trivial baselines. Across 31 firms and 58,961 pairs, aggregated per firm then averaged. Reproduced by `python3 scripts/sweep.py --firms 30`.

System	False green	Violation recall	Precision	False alarm	Abstention F1
`always_pass`	**1.000**	0.000	0.000	**0.000**	0.000
`always_abstain`	0.000	0.000	0.000	0.968	**0.632**
`always_fail`	0.000	**1.000**	0.119	0.968	0.000
`oracle`	0.000	1.000	1.000	0.000	1.000

Three readings.

The false-green rate is not redundant. `always_pass` attains a false-alarm rate of 0.000, the best possible score on compliant cases. Under any metric that rewards not crying wolf, it is indistinguishable from a careful system. It is in fact the worst possible system for deployment, reporting all 26,682 undecidable pairs as satisfied. Only the false-green rate separates them. On the anchor firm alone the separation is starker: `always_pass` reports all 201 unassessed obligations as satisfied while raising zero false alarms.

The false-green rate is also not sufficient. `always_abstain` attains 0.000 on it while detecting no violations, and reaches 0.632 abstention F1. On the anchor firm, whose expectation set declares no control pairs, it reaches 0.953 abstention F1 while being useless. Abstention scores are therefore gameable in isolation, which is why the suite must be reported jointly.

The task is solvable. The oracle reaches 1.000 on every axis. A system scoring poorly is exhibiting a property of itself, not of an ill-posed benchmark.

8.2 Language-model results

Six free models were evaluated on the declared 140-pair subset of the anchor firm under prompt v1.1.0, each under both output schemas: Nemotron 3 Ultra 550B, Nemotron 3 Super 120B and Nemotron 3.5 Lightning (NVIDIA), Laguna XS 2.1 (Poolside), Qwen 3.8 27B (Alibaba), and dots.3 Note Preview (dots-studio), the last two replacing the withdrawn

Nex models under the rule in Section 7.2. Gemma 4 31B, the other declared replacement, never passed the screen and yielded its slot to the fallback. The tables are generated from the retained replies (`scripts/fill_results.py`); the table shapes were fixed before any number existed. Every table reports coverage, because an unanswered pair cannot be a false green (Section 7).

Table 2 — main result, `abstain_offered`. One row per model, scored on the declared 140-pair subset of the anchor firm. The protocol averages across firms with a standard deviation; with one firm evaluated, each cell is that firm's value. The unparseable and answered columns guard the rest.

Model	False green	FG, answered only	Recall	Precision	False alarm	Abst. P	Abst. R	Abst. F1	Unparse.	Answered
`qwen3.8-27b`	**0.142**	0.142	0.350	0.875	0.000	0.894	0.842	0.867	0.000	140/140
`laguna-xs-2.1`	**0.125**	0.135	0.200	0.667	0.000	0.940	0.783	0.855	0.000	124/140
`dots-3-note-preview`	**0.100**	0.100	0.450	0.900	0.000	0.906	0.883	0.894	0.000	140/140
`nemotron-3-ultra-550b-a55b`	**0.067**	0.067	0.300	1.000	0.000	0.903	0.933	0.918	0.000	140/140
`nemotron-3-super-120b-a12b`	**0.050**	0.050	0.300	0.857	0.000	0.896	0.858	0.877	0.000	140/140
`nemotron-3.5-lightning`	**0.042**	0.059	0.050	0.500	0.000	0.951	0.642	0.766	0.000	92/140
`always_pass` (ref)	1.000	1.000	0.000	0.000	0.000	0.000	—	0.000	0.000	all
`oracle` (ref)	0.000	0.000	1.000	1.000	0.000	0.000	—	1.000	0.000	all

The trivial baselines are repeated in-table as reference rows, so a reader sees immediately where each model sits between the unsafe floor and the attainable ceiling. The false-alarm column is 0.000 for every row by construction, since the evaluated subset has no conforming pair (Section 9); it is kept so the table matches its declared shape.

Offered an abstention, models abstain readily and detect little. False-green rates run from 0.042 to 0.142. The lowest belongs to Lightning, which answered only 92 of 140 pairs; among the withheld pairs it actually answered its rate is 0.059. No model is close to `always_pass`, and every model is far from the oracle on violation recall, which runs from 0.050 to 0.450. Measured as rates, the missed violation is the larger failure: most of the 20 violations are missed, against a minority of the 120 withheld pairs reported as compliant. Models answer `needs_evidence` on violations whose evidence is plainly in the exports. Abstention F1 is between 0.766 and 0.918, so the abstentions mostly land on the right pairs; there are simply too many of them.

Table 2b — strict label view. Same runs, scored under the strict view (Section 6), with the deadline alerts beside them. Generated from `docs/views_results.json`.

Model	Condition	FGR, engine	FGR, strict	Recall, engine (of 20)	Recall, strict (of 10)	Alerts: fail / pass / other
`dots-3-note-preview`	offered	0.100	0.096	0.45	0.70	2 / 0 / 3
`dots-3-note-preview`	withheld	0.525	0.504	0.80	0.80	3 / 2 / 0
`nemotron-3-super-120b-a12b`	offered	0.050	0.048	0.30	0.40	2 / 0 / 3
`nemotron-3-super-120b-a12b`	withheld	0.175	0.168	0.95	0.90	5 / 0 / 0
`nemotron-3-ultra-550b-a55b`	offered	0.067	0.064	0.30	0.50	1 / 1 / 3

Model	Condition	FGR, engine	FGR, strict	Recall, engine (of 20)	Recall, strict (of 10)	Alerts: fail / pass / other
nemotron-3-ultra-550b-a55b	withheld	0.217	0.208	0.80	0.90	2 / 3 / 0
nemotron-3.5-lightning	offered	0.042	0.040	0.05	0.10	0 / 1 / 4
nemotron-3.5-lightning	withheld	0.208	0.200	0.50	0.40	3 / 0 / 2
laguna-xs-2.1	offered	0.125	0.120	0.20	0.30	1 / 0 / 4
laguna-xs-2.1	withheld	0.433	0.416	0.90	0.80	5 / 0 / 0
qwen3.8-27b	offered	0.142	0.136	0.35	0.50	2 / 0 / 3
qwen3.8-27b	withheld	0.417	0.400	0.85	0.90	3 / 2 / 0

The strict view moves false-green rates by at most 0.021 and raises recall for every model under `abstain_offered`, because the five memo pairs that most models answered `needs_evidence` stop counting as missed violations. A `pass` on a deadline days away is the one deadline-alert outcome a reviewer would never see. Offered an abstention, models gave it on 2 of 30 alerts. Without the abstention it rose to 7 of 30, three of them from Nemotron Ultra, which answered `pass` on an LCA expiring in four days.

Table 2c — audited label view. Same runs, with the two withheld pairs the blind audit overturned removed from scoring (Section 6). The scored denominator is stated in every row.

Model	Condition	Withheld pairs scored	FGR, engine	FGR, audited	FG, audited, answered only
dots-3-note-preview	offered	118 of 120	0.100	0.085	0.085
dots-3-note-preview	withheld	118 of 120	0.525	0.517	0.517
nemotron-3-super-120b-a12b	offered	118 of 120	0.050	0.034	0.034
nemotron-3-super-120b-a12b	withheld	118 of 120	0.175	0.161	0.161
nemotron-3-ultra-550b-a55b	offered	118 of 120	0.067	0.051	0.051
nemotron-3-ultra-550b-a55b	withheld	118 of 120	0.217	0.203	0.203
nemotron-3.5-lightning	offered	118 of 120	0.042	0.025	0.036
nemotron-3.5-lightning	withheld	118 of 120	0.208	0.212	0.439
laguna-xs-2.1	offered	118 of 120	0.125	0.110	0.119
laguna-xs-2.1	withheld	118 of 120	0.433	0.424	0.472
qwen3.8-27b	offered	118 of 120	0.142	0.127	0.127
qwen3.8-27b	withheld	118 of 120	0.417	0.407	0.407

Removing the two I-9 signature pairs lowers the false-green rate by 0.008 to 0.016 in 11 of the 12 runs. In the twelfth, Lightning without the abstention, it rises by 0.004, because neither removed pair was one of that run's false greens. It changes neither the ordering of models nor the direction of any condition effect. The label error the audit found is real but too small to account for the result.

Table 3 — the schema ablation. The paper's most informative experiment. Same models, both conditions, difference reported per model.

Model	FGR, abstention offered	FGR, abstention withheld	Change	Off-schema abstentions
dots-3-note-preview	0.100	0.525	+0.425	0 of 140
nemotron-3-super-120b-a12b	0.050	0.175	+0.125	0 of 140
nemotron-3-ultra-550b-a55b	0.067	0.217	+0.150	0 of 140
nemotron-3.5-lightning	0.042	0.208	+0.167	0 of 140
laguna-xs-2.1	0.125	0.433	+0.308	0 of 140
qwen3.8-27b	0.142	0.417	+0.275	0 of 140

The off-schema column counts pairs where a model returned `needs_evidence` although the schema permitted only `pass` and `fail`. A model doing this frequently is refusing a schema that cannot express what it observed, which we would report as a finding about the interface rather than an instruction-following failure.

Withholding the abstention multiplies false greens three- to fivefold. For every model with both conditions complete, the false-green rate rises when `needs_evidence` is removed from the schema, by between 0.125 (Nemotron Super) and 0.425 (dots.3), a factor of 2.94 to 5.25. No model returned an off-schema abstention: given only `pass` and `fail`, every model complied. Forced to resolve pairs it could not assess, each model split them between the two permitted answers. Most went to `fail`, an unsupported violation (32 to 99 of the 120 withheld pairs), and a substantial share went to `pass` (21 to 63). The second group is the false green a dashboard built on this schema would display. Because the prompt, sources and models are identical across conditions, the difference is attributable to the schema alone.

Table 4 — false-green rate by regime. One row per regulatory regime, one column per model. This is where the document-versus-field asymmetry in Section 9 is tested.

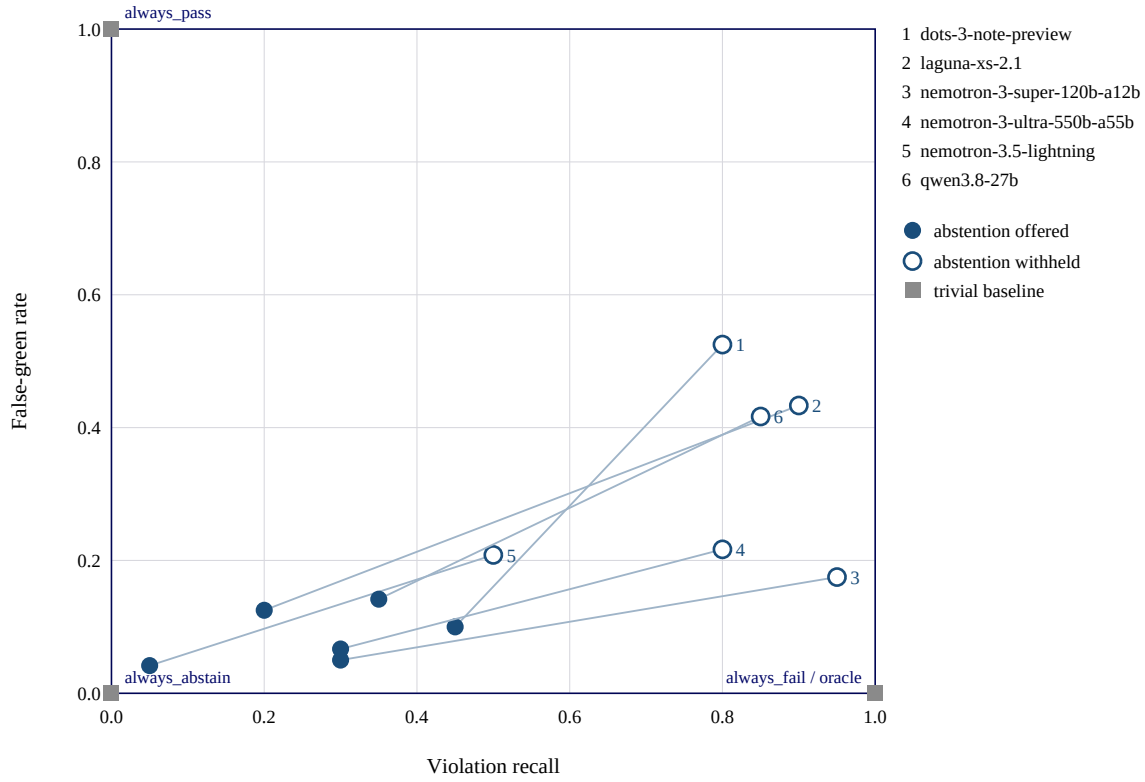
Regime	nemotron-3-ultra-550b-a55b	nemotron-3-super-120b-a12b	nemotron-3.5-lightning	laguna-xs-2.1	qwen3.8-27b	dots-3-note-preview
deadlines-penalties	0.000 (6)	0.000 (6)	0.000 (6)	0.000 (6)	0.250 (6)	0.000 (6)
fdns-sitevisit	0.000 (7)	0.000 (7)	0.000 (7)	0.000 (7)	0.000 (7)	0.000 (7)
h1b-status	0.000 (17)	0.000 (17)	0.000 (17)	0.000 (17)	0.000 (17)	0.000 (17)
i129-supplement	0.000 (8)	0.000 (8)	0.000 (8)	0.000 (8)	0.125 (8)	0.000 (8)
i9-everify	0.217 (28)	0.174 (28)	0.174 (28)	0.217 (28)	0.304 (28)	0.174 (28)
lca-dol	0.100 (33)	0.067 (33)	0.000 (33)	0.100 (33)	0.100 (33)	0.100 (33)
paf	0.000 (17)	0.000 (17)	0.059 (17)	0.000 (17)	0.000 (17)	0.000 (17)
wage-prevailing	0.000 (24)	0.000 (24)	0.000 (24)	0.389 (24)	0.278 (24)	0.278 (24)

Cells are the false-green rate with the number of scored pairs in parentheses. A dash means the regime contributed no scored pair for that model.

Table 5 — near-miss controls. For conforming cases placed exactly on the legal boundary, the rate at which each model raises an alarm, against its rate on comfortably-conforming controls. A gap indicates proximity heuristics rather than rule application. *Not produced in this study:* the evaluated subset has no conforming pairs (Section 9).

False greens are concentrated in a few regimes. Under `abstain_offered`, three regimes, I-9 and E-Verify, prevailing wage, and the LCA wage rules, hold almost every false green. Several regimes, including H-1B status and FDNS site visits, have none for any model. The regimes with false greens are the ones whose exports carry a populated column *adjacent* to the missing evidence: a signed date beside a signature, a document title beside a physical examination, an unitemized deduction total beside an itemized fee check. Section 9 returns to this.

Figure 1 — the separation plot. Violation recall on one axis, false-green rate on the other, one point per model, with the four trivial baselines marking the corners. The argument of the paper is that real systems cluster where recall looks acceptable and the false-green rate does not, and this figure either shows that or refutes it.



Each model appears twice, once per condition. Offered an abstention, the models sit low and left: few false greens, low recall. Without it they move up and right together, gaining recall at the cost of false greens. No model reaches the region of high recall and low false-green rate that the oracle occupies. Accuracy-style scoring would rank the withheld condition higher, since it detects more violations, which is the paper's central point in one picture.

What we will report if the result is null. If the models tested abstain correctly and the false-green rate is uniformly low, we will say so and the paper's claim narrows to the task, the construction method, and the instrument, with the empirical section reporting a negative result. The benchmark is publishable either way, and committing to this in advance is what makes the eventual number credible.

---

9. Analysis

Each analysis below was specified before any result existed and names the artifact that answers it, so none could be quietly dropped. Three proved not measurable on the evaluated data; they are reported as such, with the reason. All numbers come from `docs/analysis_results.json` (`scripts/analysis.py`).

Which failure dominates, per model. Error decomposed into missed violations, false greens, false alarms, and unparseable output, as a stacked proportion. A model whose error is mostly unparseable output is making a formatting claim, not a reasoning one, and must be discussed separately.

Result. Offered an abstention, models miss 6 to 14 of the 20 violations and report 5 to 17 of the 120 withheld pairs as compliant. As a rate the missed violation is the larger failure for every model. In absolute counts, false greens outnumber missed violations for Laguna, Qwen and dots.3, because withheld pairs are six times as numerous. No reply

line named a pair without a verdict, so the unparseable rate is zero throughout. The formatting failures that did occur are whole replies with no usable lines, which appear as unanswered pairs: 48 and 72 in Lightning's two runs, 16 and 12 in Laguna's. Without the abstention the picture inverts. Unsupported violations dominate (32 to 99 pairs called `fail` with no evidence either way), then false greens (21 to 63), and missed violations nearly vanish (1 to 4). The withheld condition buys its recall by calling almost everything a violation.

Where abstention is hardest. From Table 4. The hypothesis, stated before the run: obligations whose required evidence is a *field* (an I-94 expiry, a first-pay date) will show a lower false-green rate than those whose required evidence is a *document* (a posting notice, a public access file). Absence of a field is visible as an empty column; absence of a document is visible only as the absence of a record that was never expected to exist, which is a harder inference. If the asymmetry holds it is the most actionable finding for system builders, because it says where to put human review. If it does not hold, we report that the failure is uniform across evidence classes, which is also useful and which we will not suppress.

The classes are assigned mechanically, not by judgment, from each obligation's verbatim "information needed" text (`nofalsegreen/evidence_class.py`, committed before any rate was computed by class). An obligation is a *document* obligation if the text names a document, notice, posting, memo, file, list, summary, copy or clause, or a presence flag for one; otherwise it is a *field* obligation. The rule leans toward *document*, so an obligation needing both kinds of evidence counts as a document obligation. Of the 120 withheld pairs, 49 are document pairs and 71 are field pairs; the full assignment is in `docs/evidence_class_assignment.txt`.

Result: the declared test mostly holds, but the mechanism is not the one hypothesised. In 11 of the 12 runs the false-green rate on document pairs exceeds that on field pairs. The exception is Lightning without the abstention. It answered only 17 of the 49 document pairs, but the reversal holds even among pairs it answered: 0.575 on field pairs against 0.118 on document pairs. Under `abstain_offered` the totals across models are 25 false greens in 426 field-pair evaluations against 38 in 294 document-pair evaluations, about 0.06 against 0.13. The runs are not independent (the same 120 pairs, and the same models in both conditions), so we report the counts and the consistency of direction, not a p-value.

The regime breakdown (Table 4) and the excerpts below point to a narrower mechanism than "documents are harder". The false greens cluster on obligations whose required item sits *next to* a populated export column: a document-presence or signature flag beside a document title or signed date, a fee itemisation beside an unitemized deduction total. Under the mechanical rule several of these count as document obligations, because their required item is a presence flag. Where nothing adjacent is exported, as with the public access file, the six-year maximum stay or FDNS site-visit records, no model produced a single false green when abstention was offered. The actionable reading for system builders is therefore *adjacent-field substitution*. Human review belongs where an export carries a column that looks like the evidence but is not it, more than where evidence is wholly absent. This reading was formed after the results and is offered as an interpretation, not a confirmed hypothesis.

Near-miss controls. From Table 5. Distinguishes rule application from proximity heuristics. *Not measurable in the language-model study:* the evaluated subset contains only violations and withheld-evidence pairs, with no conforming pair, so there is no boundary case to probe. For the same reason the false-alarm rate reported for language models is 0.000 by construction and carries no information; tables show it as such. Both remain measurable on the generated firms, which carry conforming and near-miss pairs, and are the first analyses a funded run should add.

The schema ablation. From Table 3. A large reduction in false greens when abstention is offered locates the problem in interface design, which is an actionable finding for anyone shipping a compliance dashboard. A small reduction locates it in the models, which is a harder problem and a more important result. We commit to reporting whichever occurs.

Result. The reduction is large, so the problem sits substantially in the interface. Removing `needs_evidence` from the schema raised every model's false-green rate by 0.125 to 0.425, while the models' behaviour when the option was

available shows they can recognise most unassessable pairs. A dashboard that has no state for "could not tell" converts that recognition into green cells. This does not absolve the models. Even with the option, false-green rates of 0.04 to 0.14 remain, and recall is too low for any of them to serve as a monitor.

Evidence-pointer quality. Among pairs answered correctly, how often the cited evidence field is one ground truth records as relevant. A correct verdict reached by pointing at the wrong record is not a reliable verdict, and at scale it is indistinguishable from a guess. *Not measurable in the language-model study:* the anchor firm's labels record no ground-truth evidence fields. The cited fields are retained in every reply and are used qualitatively below.

Consistency drift. Reported if a system under test exposes headline counts. Flagged in Section 6 as the least-established axis in the suite, and it will be reported as exploratory rather than as a headline. *Not applicable here:* no model under test exposes headline counts.

Qualitative failure cases. Three to five verbatim excerpts from retained raw output showing a model asserting a verdict the records cannot support. These are the most persuasive evidence in the paper for a reader who does not care about aggregate rates, and they are available because every response is retained.

To keep these from being the most damning lines picked out of thousands, the selection rule was fixed in code before any excerpt was read (`scripts/analysis.py`). For each model, under `abstain_offered` and in the frozen subset order, the excerpt is the first `pass` on a withheld pair whose reply line cites an evidence field. One excerpt per model, quoted verbatim beside the fixed label. Applied as declared, the rule selected the same pair for four of the six models: the Section 2 signature label that the audit later overturned. We report that outcome as it fell, since four models agreeing on a mislabelled pair is itself a finding about the labels. We then apply the same rule once more, skipping overturned pairs, and mark those excerpts as the second application. This second step was added after the first application's result was seen, and is labelled as such wherever it appears.

Result, first application. For the four models whose first hit was an I-9 pair, the rule selected the same line from each: a `pass` on the Section 2 employer signature, citing `section2_signed_date`. The audit later overturned that label. Four models independently reading a populated signed-date column as evidence of a signature is the clearest evidence that this label, not the models, was wrong. The other two models' first hits stand after the audit and are given below.

Result, second application (added after the first was seen; see above). One excerpt per model, verbatim, each on a pair whose withheld label the blind audit upheld:

- Qwen 3.8 27B and dots.3 Note Preview, independently:

`h1b.wage.deductions.prohibited.business-expense … pass … pretax_deductions` (Qwen; dots.3 cites `payroll_register`). The rule asks whether any H-1B worker was charged for LCA, petition or attorney fees. The payroll export carries only an unitemized pretax deduction total, so no fee can be identified either way. Both models cite the total as evidence that no prohibited deduction was taken.

- Nemotron 3 Super 120B: `i9.remote.alternative_procedure … pass … alternative_procedure`.

The export has the alternative-procedure flag. It has nothing on E-Verify good standing, copy retention or the live-video attestation the procedure requires.

- Nemotron 3 Ultra 550B: `i9.documents.acceptable_list … pass … list_a_doc_title`. A

document title is cited as evidence that the document was acceptable. Expiry dates are blank for two of twelve forms, and the reverification documents due on 2026-06-15 are not exported.

- Nemotron 3.5 Lightning and Laguna XS 2.1, independently:

`i9.section2.physical_exam … pass … list_a_doc_title`. Physical examination of documents is inferred from the presence of a document title.

Every one is a populated column standing in for the missing evidence beside it, which is the adjacent-field substitution described above. The full reply lines are in `docs/analysis_results.json`.

---

10. Limitations

One jurisdiction, one domain. United States employment-based immigration. We claim the false-green failure generalises to other regulated domains and claim nothing about whether performance does.

The anchor firm is small and not independent. Twelve workers. Its expectations were hand-authored by the same author who built the engine it was first used to test. The generated firms address scale, not independence; a genuinely independent labelling of the anchor firm by practising immigration counsel is the obvious next step and has not been done.

The generator's violations are less diverse than the anchor firm's. It renders only mechanisms we implemented renderers for. Unrendered obligations land as withheld-evidence or not-applicable, so the generated planted-violation set is narrower than the manifest would allow.

Synthetic data does not exercise extraction. Real deployments read records from scanned documents through OCR, where a field can be invented from an adjacent label rather than merely absent. This benchmark starts from clean structured exports and therefore isolates the verdict layer, which is a deliberate scope choice and also a gap.

Verdict granularity is coarse. Real adjudication distinguishes technical from substantive defects and admits cure periods. Four states do not capture that.

Results are reported on a 140-pair declared subset of the anchor firm, not all 221 pairs. Required by the free-tier call ceiling (Section 7.2). The subset keeps every planted violation, holds the abstention-to-violation ratio at roughly 6:1 against the firm's 10:1, and covers every regime, but its smaller denominator widens the interval on every rate we report.

Model coverage is limited to free API tiers, and that limit is severe. The study was run without a compute budget, so the models evaluated are those reachable on providers' free tiers rather than a representative sample of the frontier. This is not a mild caveat. On a screening run of the 15 free chat models that met the task's context requirement, **only 6 returned a usable answer to a two-pair request**: 5 were rate-limited, 2 refused with a permissions error, 1 returned an upstream provider failure, and 1 returned an empty completion. Which models are evaluable on a given day is therefore decided by provider capacity rather than by our selection, and the model set in any free-tier result is partly an artifact of when it was run. We report the screening outcome alongside results so the composition of the model set is visible. The same instability struck during the study itself. Two of the six declared models were withdrawn from the free tier mid-run and replaced, one declared replacement never passed the screen, and one evaluated model's host cut replies off so often that its runs answered 92 and 68 of 140 pairs (Section 7.2). The final set is six models from four providers, three of them NVIDIA Nemotron variants, with nothing above roughly 550B parameters and no proprietary frontier model. Whether the largest current proprietary models exhibit the same false-green behaviour is untested, and it is the single most valuable extension of this work.

Prompts send only regime-relevant exports. Required by free per-minute token quotas (Section 7.2). Safeguarded by disclosing withheld exports in the prompt, forbidding inference about unseen data, falling back to all sources for unclassified regimes, and asserting in tests that a correct model scores identically either way. It remains a deviation from sending a firm's complete record.

The label audit is an AI's, and came after results. The blind audit (Section 6) was run by an AI agent, not by immigration counsel, and was commissioned after early results were read. Blindness to model output and coverage of every withheld pair limit how far it could be steered, and every decision is published with its reasoning, but it is not an expert review. It found 2 of 120 withheld labels contradicted by the exports. That is a lower bound on label error of that

kind, not a measurement of label quality overall: the 20 violation labels were not audited, and a counsel-led review remains the obvious next step.

No human baseline. We do not know what false-green rate a competent paralegal or attorney would attain on this task, which would be the most informative reference point of all.

---

11. Ethics and release

No real personal data. Every record is synthetic. Names are combinatorial draws from generic pools chosen to make joins legible, not to depict anyone.

Not legal advice, and not a compliance product. The rulebook is a research artifact. A high score does not establish that a system may be relied upon by an employer. We state this because the plausible misuse of this benchmark is precisely a vendor citing a score as evidence of fitness.

Not a training set. Fitting a model to these labels measures nothing, since the labels are the generator's specification.

Dual use. A benchmark measuring how often systems report unassessed obligations as satisfied could in principle guide a vendor toward suppressing the measurement rather than the failure. We judge the risk low and the transparency benefit high: the failure mode is already occurring and is currently unmeasured.

Release. Code under Apache-2.0, data and rulebook under CC-BY-4.0, with a datasheet. The harness requires only the Python standard library and runs offline.

Appendix A: Reproduction

```
python3 -m nofalsegreen.cli describe # rulebook + anchor stats
python3 -m nofalsegreen.cli score --firm northwind --baseline always_pass
python3 -m nofalsegreen.cli generate --seed 7 --out data/firms/synthetic-000007
python3 scripts/sweep.py --firms 30 # reproduces Table 1
python3 tests/test_benchmark.py # 36 assertions
```

Appendix B: Prompt and protocol

The prompts are frozen in `baselines/prompts.py` at `PROMPT_VERSION` 1.1.0 and reproduced verbatim below in both ablation conditions, since a benchmark whose prompt is not published is not reproducible.

`[INSERT: system prompt, abstain_offered]`

`[INSERT: system prompt, abstain_withheld]`

`[INSERT: one complete user prompt for a single batch, elided in the middle]`

Changing any of these text blocks changes the benchmark. Any edit bumps `PROMPT_VERSION`, and results carry the version they were produced under.